■ RAJASTHAN ROYALS AUCTION ANALYTICS

Identifying the Most Reliable Players Under Pressure

SupeRR Selector Season 2 | Round 1 Official Methodology Report & Scouting Strategy

1. Executive Summary

Modern T20 franchise cricket is decided in micro-moments of extreme leverage. Standard cricket statistics—such as raw batting averages or overall bowling economies—fail to differentiate players who produce empty runs in dead rubbers from match-winners who deliver when the game is on the line. For the Rajasthan Royals' IPL 2027 Auction cycle, this investigation builds a statistically defensible, delivery-level evaluation platform to identify and rank the **Top 10 Most Reliable Batters** and **Top 10 Most Reliable Bowlers Under Pressure**.

Analyzing **200,999 deliveries across 867 matches** from 2022 to 2026 (spanning IPL seasons 2022–2026 and international T20s), we formulate two transparent, reproducible evaluation indices: the **Batting Pressure Reliability Index (BPRI)** and the **Bowling Pressure Reliability Index (BOWPRI)**. Our framework incorporates dynamic ball-by-ball leverage tracking, baseline differential evaluation (performance under pressure relative to player normal), Empirical Bayesian shrinkage for small-sample stability, and multi-model sensitivity testing.

Core Findings:

• **#1 Batter Under Pressure: Ashutosh Sharma (BPRI: 78.22)** — Demonstrates an elite 192.83 strike rate in death overs, a +38.6 SR elevation under pressure, and clutch lower-order hitting under 12+ Required Run Rate.

• **#1 Bowler Under Pressure: Mohammad Nawaz (BOWPRI: 83.00)** — Delivers a suffocating 6.15 economy under pressure, 5.76 death economy, and 27 pressure wickets across 38.8 pressure overs.

• **Key Franchise Targets for RR:** Shashank Singh, Suryakumar Yadav, Phil Salt, Sunil Narine, and Mitchell Santner emerge as premier high-confidence anchors across distinct tactical archetypes.

2. Problem Definition & Dataset Feasibility Audit

The Round 1 mandate requires determining what counts as pressure, establishing fair comparison baselines between high-pressure and ordinary situations, and producing two uniquely ordered Top-10 lists supported by transparent mathematics.

A rigorous data audit confirmed the complete presence of delivery-level match records in ball_by_ball.csv (200,999 rows, 33 columns). Zero deliveries or player statistics were fabricated or simulated. Primary relational joins are executed strictly through match_id and player_id, guaranteeing 100% relational integrity across matches.csv, competitions.csv, and competition_team_player_mapping.csv.

Dataset	Rows	Columns	Temporal Span	Relational Role
ball_by_ball.csv	200,999	33	2022–2026	Delivery mechanics, runs, extras, wickets
matches.csv	883	22	2022–2026	Match metadata, targets, knockouts
competitions.csv	141	4	IPL & T20I	Competition format and season taxonomy
mapping.csv	6,413	6	Universal	Official player_id & team_id linkage

3. Transparent Pressure Classification Engine

A foundational analytical principle of this investigation is that **not every death-over ball is a pressure situation**. For example, when a chasing team requires 15 runs off 35 balls with 8 wickets in hand, the Required Run Rate is 2.57; the batting and bowling units face virtually zero pressure despite operating in overs 15–20. Conversely, a batter entering at 28/3 in the 4th over faces enormous collapse-recovery pressure.

Our dynamic engine evaluates delivery context using a continuous 0–100 score partitioned into four validated tiers:
• **Low Pressure (<35):** 23.01% of balls (comfortable run rates, ample wickets, decided contests).
• **Moderate Pressure (35–59):** 31.99% of balls (standard par-scoring middle-over phases).
• **High Pressure (60–79):** 30.39% of balls (death-over acceleration, steep chases RRR 10–12, early collapses).
• **Extreme Pressure (>=80):** 14.61% of balls (RRR > 13.0, close finishes balls <= 18 with runs needed <= 30, super overs, knockout eliminators).

Mathematical Modeling of Chase Leverage:
In Innings 2, pressure is calculated continuously via a logistic sigmoid of Required Run Rate modulated by wicket strain and endgame proximity:
*P_chase = [100 / (1 + exp(-0.45 * (RRR - 9.0)))] * [1 + max(0, (5 - Wickets_in_Hand) * 0.08)] + Knockout_Boost*

4. Performance Methodology & Index Architecture

To ensure fair, role-sensitive player ratings, we reject raw counting statistics (which bias toward players facing high volume) in favor of multidimensional relative efficiency. Performance under High/Extreme pressure is evaluated across six structured pillars:

Evaluation Pillar	Batting (BPRI) Weight	Bowling (BOWPRI) Weight	Strategic Rationale & Metrics
1. Baseline Differential	30%	30%	Measures lift under pressure: (Pressure SR - Normal SR) & (Normal Econ - Pressure Econ).
2. Primary Efficiency	20%	20%	Batting Pressure Strike Rate / Bowling Pressure Economy Rate (inverted).
3. Control & Prevention	15%	15%	Batting: Boundary % & Dot % suppression / Bowling: Dot % & Boundary conceded % suppression.
4. Chase & Situational	15%	15%	Batting: 2nd innings chase SR & Avg / Bowling: Wicket-taking strike rate under pressure.
5. Death-Over Mastery	10%	10%	Overs 16–20 Strike Rate & Boundaries / Overs 16–20 Economy & Wickets per 100 balls.
6. Consistency Factor	10%	10%	Inverted standard deviation of match-by-match performance under pressure.

Statistical Reliability & Empirical Bayesian Shrinkage:
To avoid sample size distortion (e.g. a player with 15 pressure balls having an inflated 220 SR), we enforce strict qualification thresholds (Batters: >=10 innings, >=75 pressure balls; Bowlers: >=10 innings, >=60 pressure balls). All raw ratings undergo Empirical Bayesian shrinkage toward the league prior mean:
*Index_final = [N / (N + K)] * Index_raw + [K / (N + K)] * Prior_Mean* (where K_bat=60, K_bowl=50). Confidence indicators (40% to 99%) are reported transparently for every player.

5. Top 10 Most Reliable Batters Under Pressure

Below is the official ordered ranking of the Top 10 Batters evaluated across all eligible modern T20 records. Rank 1 represents the highest-rated pressure performer.

Rank	Player Name (ID)	Team / Role	HP Balls	HP Avg	HP SR	Death SR	BPRI	Conf.
#1	**Ashutosh Sharma** ID: 8101	Delhi Capitals **Lower-order Power Hitter**	341	25.0	**168.6**	192.8	**78.22**	99%
#2	**Shashank Singh** ID: 2756	Punjab Kings **Finisher**	492	42.0	**170.5**	195.4	**77.73**	99%
#3	**Suryakumar Yadav** ID: 1699	India **Middle-order Aggressor**	1005	41.7	**182.7**	217.2	**76.90**	99%
#4	**Tom Banton** ID: 6273	England **Middle-order Aggressor**	114	42.6	**186.8**	188.6	**74.05**	46%
#5	**Reeza Hendricks** ID: 768	South Africa **Aggressive Opener**	365	44.2	**169.6**	178.0	**72.93**	99%
#6	**Naman Dhir** ID: 37052	India **Middle-order Aggressor**	342	29.0	**186.3**	200.6	**72.91**	99%
#7	**Quinton de Kock** ID: 2101	Mumbai Indians **Aggressive Opener**	455	38.8	**170.3**	232.0	**72.20**	99%
#8	**Phil Salt** ID: 5553	England **Aggressive Opener**	292	36.5	**187.3**	208.4	**72.12**	99%
#9	**Rishabh Pant** ID: 4703	Lucknow Super Giants **Middle-order Aggressor**	473	29.8	**176.5**	200.8	**71.31**	99%
#10	**Tim David** ID: 6396	Australia **Finisher**	937	35.4	**181.3**	200.9	**71.28**	99%

6. Top 10 Most Reliable Bowlers Under Pressure

Below is the official ordered ranking of the Top 10 Bowlers evaluated across all eligible modern T20 records. Rank 1 represents the highest-rated pressure performer.

Rank	Player Name (ID)	Team / Role	HP Overs	HP Econ	Death Econ	HP Wkts	BOWPRI	Conf.
#1	**Mohammad Nawaz** ID: 2490	Pakistan **Middle-over Spin Controller**	38.8	**6.15**	5.76	27	**83.00**	99%
#2	**Abrar Ahmed** ID: 5979	Pakistan **Middle-over Spin Controller**	40.8	**6.51**	6.88	19	**77.96**	99%
#3	**Matthew Humphreys** ID: 22471	Ireland **Powerplay Specialist (Spin)**	13.8	**6.80**	4.97	10	**76.68**	42%
#4	**George Linde** ID: 4229	South Africa **Middle-over Spin Controller**	13.0	**7.38**	6.32	7	**76.10**	40%
#5	**Sunil Narine** ID: 781	Kolkata Knight Riders **Death-over Specialist (Spin)**	97.7	**7.27**	6.06	34	**75.71**	99%
#6	**Imad Wasim** ID: 3578	Pakistan **Middle-over Spin Controller**	17.3	**6.00**	7.67	8	**75.62**	52%
#7	**Corbin Bosch** ID: 3629	Mumbai Indians **Death-over Specialist (Pace)**	21.0	**7.14**	8.77	10	**74.56**	63%

#8	**Hasan Mahmud** ID: 6643	Bangladesh **Middle-over Wicket-taker (Pace)**	20.0	**7.50**	8.39	11	**74.09**	60%
#9	**Sufiyan Muqeem** ID: 34485	Pakistan **Middle-over Spin Controller**	24.3	**4.89**	7.11	9	**73.81**	73%
#10	**Mitchell Santner** ID: 4101	New Zealand **Death-over Specialist (Spin)**	95.5	**7.21**	9.34	38	**73.53**	99%

7. Multi-Model Sensitivity Analysis

To satisfy the competitive challenge mandate regarding weight transparency, we conducted sensitivity experiments across 4 distinct alternative weighting schemes (Baseline-first, Death-overs focus, Finishing focus, and Wicket-taking aggression).

• **Batting Spearman Rank Correlation: 0.977 to 0.993** across all configurations.

• **Bowling Spearman Rank Correlation: 0.978 to 0.989** across all configurations.

Conclusion: The generated rankings are robust, objective, and mathematically resilient against arbitrary subjective weight manipulation.

8. Rajasthan Royals IPL 2027 Auction Acquisition Strategy

Translating statistical pressure reliability into franchise roster architecture at the Sawai Mansingh Stadium (Jaipur). Below are actionable squad-building blueprints tailored to RR's tactical needs:

A. Domestic Lower-Order Finishing Chasm: Ashutosh Sharma & Shashank Singh

• **Franchise Need:** Rajasthan Royals have historically suffered when their top-3 fail to bat through 16 overs. In tight chases requiring 45 off 18 balls at Jaipur, lower-order boundary striking has been an Achilles heel.
• **Tactical Fit:** Ashutosh Sharma (BPRI: 78.22, Death SR: 192.83) and Shashank Singh (BPRI: 77.73, HP Avg: 41.95) provide elite domestic Indian finishing power. Ashutosh's ability to sweep fast bowlers and target square boundaries neutralizes defensive death-bowling lines, while Shashank offers an ice-cool collapse stabilizer.

B. Marquee Anchor & Powerplay Dynamic: Phil Salt & Suryakumar Yadav

• **Franchise Need:** Maintaining relentless scoring momentum without stalling when an early wicket falls in the powerplay.
• **Tactical Fit:** Phil Salt (HP SR: 187.33) complements Yashasvi Jaiswal by dismantling opening pace attacks, creating powerplay scores of 65+ regularly. Suryakumar Yadav (BPRI: 76.90 over 1,005 pressure balls) remains the premier 360-degree match-winner in cricket history, transforming middle overs (7–15) into an offensive launchpad.

C. All-Phase Suffocating Spin Control: Sunil Narine & Mitchell Santner

• **Franchise Need:** Defending sub-165 scores on dry, two-paced Jaipur wickets where pace bowling can be vulnerable to flat batting tracks.
• **Tactical Fit:** Sunil Narine (6.06 death economy, 97.66 pressure overs) and Mitchell Santner (7.21 pressure economy, 38 pressure wickets) guarantee 8 overs of boundary denial. Santner represents an exceptional value overseas leadership acquisition who can bowl in powerplay, middle, and death phases.

D. Analytical Archetypes for Scouting (Mohammad Nawaz, Abrar Ahmed)

• **Franchise Insight:** While Pakistani players are currently ineligible for direct IPL auction participation, their dominance in our mathematical rankings (Nawaz #1, Abrar #2) provides RR's global analytics scouts with the **ideal statistical archetype**: left-arm dart spinners who bowl flat lengths into the pitch and mystery leg-spinners who force false shots through carrom variation.

9. Analytical Limitations & Governance

1. **Venue & Pitch Heterogeneity:** True bounce and boundary dimensions vary across venues (e.g. Wankhede vs Chepauk vs Jaipur). Incorporating ball-tracking tracking trajectory data in Round 2 will enhance pitch-adjusted valuations.
2. **Matchup Sample Sizes:** Specific head-to-head batter vs bowler encounters under pressure have limited samples; hierarchical Bayesian models provide regularization.
3. **Impact Player Evolution:** The IPL Impact Player rule has shifted middle-order batting depth since 2023, slightly lowering batting risk thresholds.

10. Conclusion

This submission delivers a complete, statistically verified, and strategically actionable scouting system for Rajasthan Royals. By combining ball-by-ball leverage calculation, baseline differential comparisons, Bayesian shrinkage, and auction role modeling, we have identified the players who deliver when it matters most. The resulting top-10 lists provide Rajasthan Royals with the analytical edge required to dominate the IPL 2027 Auction.